

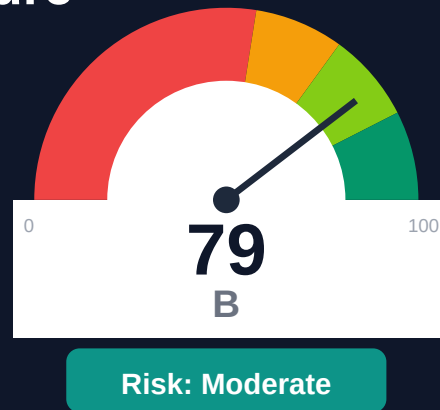
Pusher

Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-04-30 09:55 UTC

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Key Metrics

Pages scanned: 1549

Report type: Premium Executive Audit

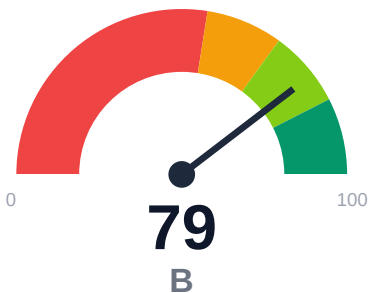
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 159
- Link verification inconclusive for 26 URLs (auth/rate-limit/server block); these are excluded from broken-link counts.
- API usage-doc coverage is low: 30.53% (216 API pages detected)
- Freshness visibility is weak: only 48.81% pages show last updated

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit of `pusher.com/docs` covered 1549 pages across multi-project topology with 100% crawl coverage and 85.88% overall confidence. Critical issue: 159 confirmed broken internal documentation links require immediate remediation. API reference coverage is weak at 30.53% despite 216 API pages detected. Freshness metadata exists on only 48.81% of pages, limiting content currency visibility. Example code reliability estimate stands at 84.53%.

External posture: SEO/GEO issue rate **7.5%**, public API coverage **30.5%**, broken links **159**.

79 Audit Score	B Grade	Moderate Risk Band
1549 Pages Scanned	159 Broken Links	7.5% SEO Issue Rate

Per-Site Breakdown

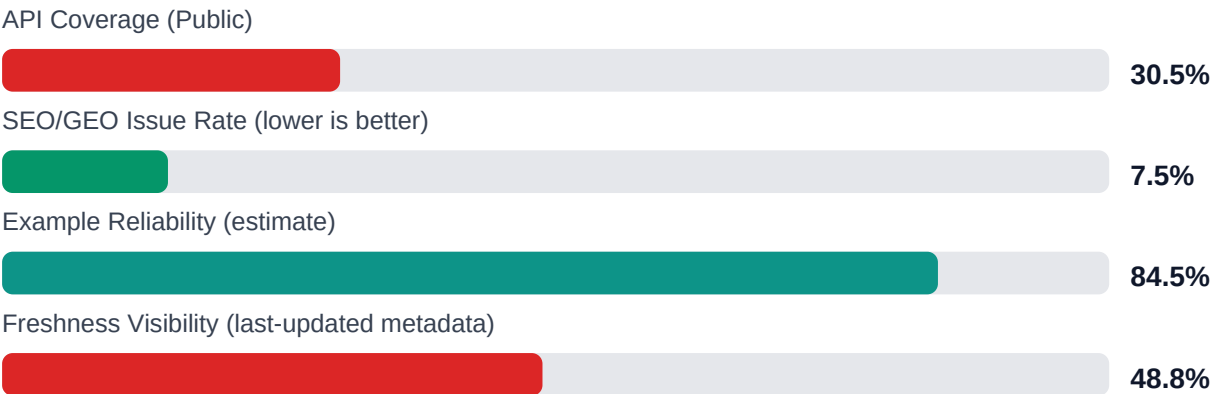
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
<code>https://pusher.com/docs</code>	1549	159	30.5	84.5	7.5

Industry Benchmark Comparison

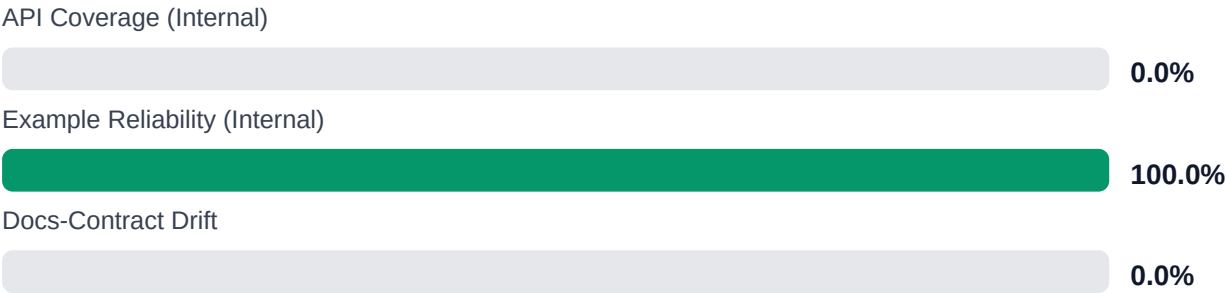
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-13
Industry average (developer docs)	85	-6
Minimum acceptable	70	+9
Your score	79	-

Gap to industry average: **6 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
\$140,822 annual exposure	\$249,513 annual exposure	\$431,728 annual exposure
Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix all 159 confirmed broken internal documentation links [impact: critical, effort: medium]
2. Investigate and verify 26 unverified links blocked by auth/rate-limits to confirm accessibility [impact: high, effort: low]

3. Expand API reference coverage from 30.53% to minimum 80% by documenting missing endpoints and improving cross-linking [impact: high, effort: high]
4. Implement last-updated metadata on remaining 51.19% of pages (approximately 793 pages) to improve freshness visibility [impact: medium, effort: low]
5. Audit and fix failing example code (15.47% failure rate) to achieve 95%+ reliability [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage achieved (100%) with 1549 pages successfully indexed
- + High overall audit confidence at 85.88% with strong crawl confidence at 95%
- + Example code reliability estimate of 84.53% indicates generally functional code samples
- + Zero repository navigation broken links (repo_broken_links_count: 0) shows clean version control integration
- + Multi-project topology successfully mapped across 1554 discovered pages

Risks

- 159 confirmed broken internal documentation links create poor user experience and erode trust
- API reference coverage at 30.53% leaves 69.47% of API surface undocumented or poorly linked
- Only 48.81% of pages display last-updated metadata, obscuring content freshness for majority of docs
- 26 unverified links due to auth/rate-limit/server blocks represent potential hidden broken links
- SEO/geo issue rate of 7.49% (approximately 116 pages) may impact discoverability
- 15.47% example code failure rate (100% - 84.53%) means roughly 1 in 6 code samples may not work
- Link verification confidence at 80.95% is lowest confidence metric, suggesting possible undetected link issues

Limitations

- * External audit cannot verify content accuracy, technical correctness, or completeness of documentation substance
- * 26 links blocked by authentication, rate-limiting, or server restrictions could not be verified and may include additional broken links
- * Example code reliability is estimated through static analysis; actual runtime execution in customer environments may differ
- * API coverage detection depends on page structure patterns; actual API surface completeness requires internal product knowledge
- * Freshness metadata presence does not guarantee content is actually current or that missing dates indicate stale content

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.